

Properties of Ratio

1. If we multiply or divide the numerator and denominator of a ratio by the same number, the ratio does not change.

Example: $3 / 4 = 0.75$

$3 * 2 / 4 * 2 = 6 / 8 = 0.75$ (Numerator and denominator multiplied by 2, and there is no change in ratio)

Ratio remains unchanged if both antecedent and consequent are multiplied by same number.



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$$\frac{3}{4} = 0.75$$

$$\frac{\cancel{6}}{\cancel{8}} \frac{3}{4} = \underline{0.75}$$

$$\frac{3 \times 2}{4 \times 2} = \frac{6}{8}$$

$$= \underline{\underline{0.75}}$$

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Que: If $p:q = r:s = t:u = 2:3$ then what is the value of $(p+r+t) / (q+s+u)$?

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$$\frac{p}{q} = \frac{r}{s} = \frac{t}{u} = \frac{2}{3}$$

$$\frac{p}{q} = \frac{2}{3}$$

$$p = \frac{2q}{3}$$

$$\frac{r}{s} = \frac{2}{3}$$

$$r = \frac{2s}{3}$$

$$\frac{t}{u} = \frac{2}{3}$$

$$t = \frac{2u}{3}$$

$$\frac{p+r+t}{q+s+u} = \frac{2}{3}$$
$$\frac{\frac{2q}{3} + \frac{2s}{3} + \frac{2u}{3}}{q+s+u}$$

$$\frac{p+r+t}{q+s+u} = \frac{2}{3}$$

$$\frac{2}{3} (q+s+u) = \frac{2}{3}$$

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$$\frac{p}{q} = \frac{r}{s} = \frac{t}{u} = \frac{2}{3}$$

$$\frac{p+r}{q+s} = \frac{t}{u} = \frac{2}{3}$$

$$\frac{p+2}{q+3} = \frac{r}{s} = \frac{t}{u} \quad \frac{p+r+2}{q+s+3} = \frac{t}{u}$$

$$\frac{p+t}{q+u} = \frac{r}{s} = \frac{2}{3}$$

Que: If $p:q = r:s = t:u = 2:3$ then what is the value of $(mp+nr+ot) : (mq+ns+ou)$?

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$$\frac{mp}{mq} = \frac{nr}{ns} = \frac{ot}{ou} = \frac{2}{3}$$

$$\frac{mp+nr+ot}{mq+ns+ou} = \frac{2}{3}$$

$$\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{2}{3}$$

$$\frac{a+c+e}{b+d+f} = \frac{2}{3}$$

$$\frac{mp+nr+ot}{mq+ns+ou} = \frac{2}{3}$$

$$\begin{aligned} a &= mp \\ b &= mq \\ c &= nr \\ d &= ns \\ e &= ot \\ f &= ou \end{aligned}$$

Properties of Proportion

1. If four numbers are in proportion then product of extremes is equal to the product of means.

Example: Let a, b, c and d be in proportion, then we can clearly write $a:b :: c:d$, that is $a/b = c/d$, that is **$ad = bc$** , where a and d are the extremes of this proportion and b and c are the means of this proportion.

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extreme
 $a:b :: c:d$
mean

$a/b | c/d$

$$\frac{a}{b} = \frac{c}{d}$$

$$ad = bc$$

Properties of Proportion

2. If three numbers (say a, b, c) are in continued proportion, then

a) Mean Proportion = $\sqrt{ac}$

b) Third Proportion = b^2 / a

If three numbers (say a, b, c) are in continued proportion, then

a is called as first proportion

b is called as mean proportion

c is called as third proportion

Properties of Proportion

2. If three numbers (say a, b, c) are in continued proportion, then

a) Mean Proportion = $\sqrt{ac}$

b) Third Proportion = b^2 / a

a, b, b, c

If three numbers (say a, b, c) are in continued proportion, then

a is called as first proportion ✓

b is called as mean proportion / second proportion

c is called as third proportion ✓

$$b^2 = ac$$
$$c = \frac{b^2}{a}$$

a, b, b, c

a, b, c

$\frac{a}{b} = \frac{b}{c}$

$b^2 = ac$

$b = \sqrt{ac}$

$a:b :: b:c$

$b \cdot b = a \cdot c$

$b^2 = ac$

a, b, c, d

$$\frac{a}{b} = \frac{c}{d}$$

$$a:b :: c:d$$

Que: Find the mean proportion between 5 and 45.



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Que: Find the mean proportion between 5 and 45.

→ 15

$$\text{mean proportion} = \sqrt{ac}$$

$$= \sqrt{5 \times 45}$$

$$= \sqrt{5 \times 9 \times 5}$$

$$= \sqrt{5^2 \times 3^2}$$

$$= 5 \times 3$$

$$= 15$$

Que: Find the mean proportion between 5 and 45.

$$5, x, 45$$

$$\frac{5}{x} = \frac{x}{45}$$

$$x^2 = 5 \times 45$$

$$x = \sqrt{5 \times 45}$$
$$= \underline{\underline{15}}$$